MATH2241 – Introduction to mathematical analysis (1st semester 2526)

Tutorial 1 Solution

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1. Suppose that a, b, ϵ are in an ordered field F and $a < b + \epsilon$ for any $\epsilon > 0$. Prove that $a \leq b$.

   Solution.
   Suppose $b < a$, then take $\epsilon = a - b > 0$, we have

   $$a < b + (a - b)$$
   $$a < a \qquad !!!$$

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2. For each of the following sets, find the supremum and infimum (without proof) if they exist. State also whether the supremum and infimum are the maximum and minimum respectively.

   (a) $S_1 = \{x \in \mathbb{R} : x \neq 0, \frac{1}{x} > 2\}$
   (b) $S_2 = \{\frac{1}{k} + \frac{2}{m} + \frac{3}{n} : k, m, n \in \mathbb{N}\}$
   (c) $S_3 = \{3^x : x \in \mathbb{Z}\}$

   Solution.

   (a) $\sup S_1 = \frac{1}{2} \neq \max S_1$, $\inf S_1 = 0 \neq \min S_1$.
   (b) $\sup S_2 = 6 = \max S_2$, $\inf S_2 = 0 \neq \min S_2$.
   (c) $\sup S_3 = \infty \neq \max S_3$, $\inf S_3 = 0 \neq \min S_3$.

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3. Let A be a nonempty bounded subset of R. What can we say about A if inf A = sup A?

   Solution.
   If A contains two numbers a, b and $a < b$, then $\inf A \leq a < b \leq \sup A$, contradiction. A is a single element set.

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4. Let A and B be nonempty bounded subsets of $\mathbb{R}$. Prove that if $A \subseteq B$, then

   $$\inf B \leq \inf A \leq \sup A \leq \sup B$$

Solution.
If $\inf B > \inf A$, let $r = \inf B - \inf A > 0$, there is $w \in A$ such that $w - \inf A < r \implies w < \inf B \implies w \notin B$, contradiction. The last inequality is similar, the middle one is by definition. ■

5. Let A and B be nonempty bounded subsets of $\mathbb{R}$. Define

$$A + B = \{a + b : a \in A \text{ and } b \in B\}$$
$$AB = \{a \times b : a \in A \text{ and } b \in B\}$$

(a) Is that $\sup(A + B) = \sup A + \sup B$?

(b) Is that $\sup(AB) = (\sup A) \times (\sup B)$?

Solution.

(a) $\forall x \in A + B$, $x = a + b \leq \sup A + \sup B$ for some $a \in A$ and $b \in B$. Now $\sup A + \sup B$ is an upper bound of $A+B$. If $\sup(A+B) < \sup A+\sup B$, take $r = \sup A+\sup B-\sup(A+B) > 0$, there are $a' \in A, b' \in B$ such that $\sup A - a' < r/2$ and $\sup B - b' < r/2$, then $\sup(A + B) < a' + b' \in A + B$, contradiction.

(b) Let $A = \{1, 2\}$, $B = \{-2, -1\}$, then $AB = \{-4, -2, -1\}$. It's not true.

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6. Let A be a nonempty subset of $\mathbb{R}$ and let w be an upper bound of A. Suppose that for every $n \in \mathbb{N}$, there exists $a_n \in A$ such that $w - \frac{1}{n} \leq a_n$. Prove that $w = \sup A$.

Solution.
If $r = w - \sup A > 0$, there is $n \in \mathbb{N}$ such that $r > \frac{1}{n} > 0$, then there is $a_n \in A$ such that $w - \frac{1}{n} \leq a_n \implies$

$$\begin{aligned} a_n &\geq w - \frac{1}{n} \\ &\geq w - r \\ &= \sup A \qquad !!! \end{aligned}$$

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7. (a) Give an example of a set of irrational numbers whose supremum is rational.

(b) Give an example of a set of rational numbers whose supremum is irrational.

Solution.

(a) $S_i = \{\frac{3}{\pi}, \frac{3.1}{\pi}, \frac{3.14}{\pi}, \frac{3.141}{\pi}, ...\}$, $\sup S_i = 1$

(b) $S_r = \{3, 3.1, 3.14, ...\}$, $\sup S_r = \pi$

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8. Show that

   (a) the root $\sqrt[m]{n}$ for $m, n \in \mathbb{N}$ is either an integer or an irrational number.

   (b) $\sqrt{n} + \sqrt{n+1}$ is an irrational number for every $n \in \mathbb{N}$.

   (c) $\sqrt{n + \sqrt{n}}$ is an irrational number for every $n \in \mathbb{N}$.

   Solution.

   (a) If not, $\sqrt[m]{n} = \frac{p}{q}$, where p and q are integers with largest common divisor 1 and $|q| \neq 1$. Then $n = \frac{p^m}{q^m}$, which implies p and q have a common divisor other than 1, a contradiction.

   (b) $n^2 < n(n+1) < (n+1)^2$, hence $n(n+1)$ is not a complete square. Then $\sqrt{n(n+1)}$ is an irrational number by (a). Now,

   $$(\sqrt{n} + \sqrt{n+1})^2 = 2n + 1 + 2\sqrt{n(n+1)},$$

   it follows that $(\sqrt{n} + \sqrt{n+1})^2$ is irrational and hence so is $\sqrt{n} + \sqrt{n+1}$.

   (c) If $n = k^2$ for some $k \in \mathbb{N}$, then $\sqrt{n + \sqrt{n}} = \sqrt{b^2 + b} = \sqrt{b(b+1)}$, it is not rational. If n is not a square of an integer, then $\sqrt{n}$ is irrational and $\sqrt{n + \sqrt{n}}$ is also irrational.

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